CodeZone Research Group at NADI 2026 Shared Task: Cross-Domain Score Calibration for Spoken Arabic Dialect Identification

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Abstract

We describe the CodeZone Research Group system for NADI 2026 Shared Task 2, spoken Arabic dialect identification over Modern Standard Arabic and 19 country-level dialects under cross-domain evaluation. Our system ensembles an ECAPA-TDNN language identifier fine-tuned on the shared task data with a Whisper-large-v3 dialect classifier that we further adapt using waveform- and spectrogram-level augmentation, fusing their log-softmax scores. This ensemble reaches 94.35% accuracy on in-domain validation but initially retained only 25.40% on the official blind test, a collapse that affected every participating team. We show that about a fifth of this gap is a calibration failure rather than a modelling failure: discarding per-class bias terms fit in domain raised official accuracy from 0.2540 to 0.3918 and lowered the shared task's average cost from 0.2077 to 0.1515, and a softening temperature reduced cost further to 0.1469; neither change touched either model. Our submission ranked 10th of 16 teams, and we show analytically why uniform additive score shifts cannot affect this metric while temperature scaling can.

1 Introduction

Arabic is a continuum of regional varieties that differ in phonology, lexicon and morphology, and that coexist with Modern Standard Arabic (MSA) in a diglossic relationship. Identifying which variety a speaker is using conditions model selection for speech recognition and supplies labels for varieties that MSA-centric systems serve poorly. The spoken setting is harder than the textual one: orthographic cues are unavailable, the evidence is acoustic and prosodic and often carried by a few short segments, and dialect identity is entangled with speaker and channel, so a model can easily learn the recording environment instead of the dialect.

NADI 2026 Subtask 2 (Sullivan et al., 2026) is the second edition of the multidialectal Arabic speech processing shared task (Talafha et al., 2025) and the sixth NADI campaign overall (Abdul-Mageed et al., 2020, 2021, 2022, 2023, 2024). Systems assign each utterance one of 20 labels — MSA plus 19 country-level dialects — ranked by accuracy with an LRE-style average detection cost as secondary metric. Training data are broadcast-style while the blind test comes from other acoustic domains, making this a cross-domain robustness test rather than a held-out classification test. That distinction dominated our development.

Our system combines SpeechBrain's ECAPA-TDNN language identifier (Desplanques et al., 2020), pretrained on VoxLingua107 (Valk and Alumäe, 2021) and fine-tuned by us, with a Whisper-large-v3 encoder (Radford et al., 2022) plus attention pooling released by Elleuch et al. (2025a), which we adapt using augmentation; we fuse their log-softmax scores and calibrate before decoding.

Our central finding is that the only change that measurably improved our out-of-domain score acted at the decision layer, not inside either model. The ensemble reached 94.35% in-domain validation accuracy but 0.2540 on the official test, with a low false-alarm rate (0.025) and a miss rate seven times higher (0.183), under-confident rather than misconfigured. Discarding the per-class bias fit in domain raised official accuracy to 0.3918, a gain of 13.8 points, and cut cost from 0.2077 to 0.1515; softening the temperature then reduced cost to 0.1469. Neither change touched either model, though the accuracy recovered is only about a fifth of the gap between our in-domain and out-of-domain scores. We placed 10th of 16 teams. We further show this cost metric is provably invariant to uniform additive score shifts, worth knowing before spending a limited submission budget on

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them. Our code is available.[1]

2 Background

2.1 Task Setup

Subtask 2 is a closed-set problem. The input is a single-channel utterance at 16 kHz with no transcript, speaker identity or metadata. The output is a hard label over 20 classes *and* the full vector of 20 scores that produced it, because the secondary metric reads ratios between class scores rather than the $\arg\max$ alone. That metric treats each dialect as an independent binary detector, converts scores to log-likelihood ratios, thresholds at zero, and sums the false-alarm and miss rates averaged over the 20 classes. Every class contributes equally regardless of frequency, and the metric depends on the *calibration* of the scores, not only their ranking. We entered Subtask 2 alone, with one system, out of the five tasks NADI 2026 spans.

2.2 Dataset

The data descend from ADI-17, released for the MGB-5 challenge (Ali et al., 2019), which collected roughly 3,000 hours of dialectal Arabic speech from YouTube across 17 Arabic-speaking countries. Elleuch et al. (2025a) extended it into ADI-20: 3,556 hours over 19 dialects plus MSA, covering every Arabic-speaking country. NADI 2026 distributes ADI-20 Micro, a stratified subset of about ten hours per dialect — 67,015 training and 10,806 validation utterances of broadcast-style YouTube material (news, talk shows, interviews, entertainment) — so the training distribution carries that medium's channel and production characteristics. The blind test contains 878 unlabelled utterances from other acoustic domains.

2.3 Related Work

Earlier NADI campaigns (Abdul-Mageed et al., 2020, 2021, 2022, 2023, 2024) addressed dialect identification from *text*, where orthographic cues are directly observable; the speech setting was introduced in the previous edition (Talafha et al., 2025). Spoken ADI has developed along two lines: small controlled corpora, where Alrehaili et al. (2023) classify eight dialects from 672 recordings with a convolutional network at 83% accuracy, in domain and over a coarser label set; and large-scale work from ADI-17 to ADI-20 (Elleuch et al., 2025a), where the strongest systems fine-tune large pretrained encoders and the best NADI 2025 system (Elleuch et al., 2025b) paired a Whisper encoder with attention pooling to reach 79.83% accuracy.

Both lines optimise the model and report in-domain accuracy. We work *after* the model, asking what happens at the decision layer when the evaluation distribution changes. Temperature scaling is established as an in-domain calibration technique (Guo et al., 2017); we show that under domain shift its direction reverses — softening rather than sharpening helps — and that a per-class bias fit in domain transfers negatively, degrading out-of-domain accuracy rather than merely failing to improve it.

3 System Description

Two independent classifiers produce class scores, a fusion stage combines them, and a decision stage calibrates before thresholding. Training details are in Section 4 and pseudocode in Appendix A.

Branch A. ECAPA-TDNN (Desplanques et al., 2020) from SpeechBrain (Ravanelli et al., 2024), initialised from the VoxLingua107 checkpoint (Valk and Alumäe, 2021) and re-headed with a fresh 20-way linear layer. Attentive statistics pooling is a strong inductive bias for utterance-level identity decisions, and pretraining across 107 languages supplies features already tuned to language-discriminative variation. We do not take this to imply invariance to nuisance factors: probing an end-to-end Arabic dialect identification model, Chowdhury et al. (2020) report speaker-invariant representations but find that gender and channel information is retained throughout the network, which is consistent with the channel sensitivity we observe out of domain (Section 5). The clearer limitation of this branch is that VoxLingua107 treats Arabic as one class, so nothing in that pretraining required within-Arabic discrimination.

Branch B. This branch starts from a model with real dialectal supervision: the released `ADI-whisper-ADI20` checkpoint (Elleuch et al., 2025a), a Whisper-large-v3 encoder (Radford et al., 2022) with attention pooling and a linear head, trained on a 53-hour-per-dialect subset of ADI-20. Its positional embeddings are sized for a fixed 30-second window, so every utterance is padded or trimmed to 480,000 samples before log-mel extraction, whereas Branch A takes variable-length in-

[1] https://github.com/SarahYassine/NADI-2026-Task-2-Spoken-Dialect-Identification

put; the branches therefore see genuinely different views of the same audio, which is what makes fusing them worthwhile. From the submission ordering, we permute output columns rather than retrain the head. Branch B therefore brings roughly 1,060 hours of ADI-20 audio from outside the shared task release, as Branch A brings VoxLingua107 pretraining. Both are permitted: Subtask 2 is an open-data task, and neither resource overlaps the blind test. We did not download or train on ADI-17 or ADI-20 ourselves and used no external noise corpora. Our scores should nonetheless be read as those of an externally pretrained system rather than one trained only on the released ADI-20 Micro split, and the comparison in Section 5 between S1 and S2 is largely a comparison of what the two pretraining corpora contain.

Fusion. With $\mathbf{z}^A, \mathbf{z}^B \in \mathbb{R}^{20}$ the branch outputs and σ the softmax, we fuse in log-probability space as $\mathbf{s} = w \log \sigma(\mathbf{z}^B) + (1-w) \log \sigma(\mathbf{z}^A)$. Fusing after the softmax puts both branches on a common bounded scale, so the result does not depend on either model's arbitrary output magnitude.

Decision stage. The metric converts scores into per-class log-likelihood ratios,

$$\mathrm{LLR}_i(\mathbf{s}) = -\log \frac{1}{C-1} \sum_{j \neq i} \exp(s_j - s_i), \quad (1)$$

with $C = 20$; class i is detected when $\mathrm{LLR}_i \geq 0$, the Bayes threshold for P_{target}=0.5, C_{miss}=C_{fa}=1. We calibrate by $\tilde{\mathbf{s}} = \mathbf{s}/T + \mathbf{b}$ before Eq. 1. Three properties of Eq. 1 fixed our choice of T and $\mathbf{b}$; proofs are in Appendix B.

Proposition 1. *For any $c \in \mathbb{R}$ and $T > 0$: (i) $\mathrm{LLR}_i(\mathbf{s}+c\mathbf{1}) = \mathrm{LLR}_i(\mathbf{s})$; (ii) if $i^\star = \arg\max_j s_j$ then $\mathrm{LLR}_{i^\star}(\mathbf{s}/T) \geq 0$; (iii) $\arg\max_j s_j/T = \arg\max_j s_j$, and $\mathrm{LLR}_i(\mathbf{s}/T) \to 0$ as $T \to \infty$.*

Part (i) rules out a uniform bias as a lever, however tempting one is when misses dominate false alarms. Parts (ii) and (iii) say temperature can never change accuracy, and that raising T compresses all ratios toward zero. Since the top class is detected at any temperature, misses come only from utterances whose true dialect is not ranked first; for those the true class has $\mathrm{LLR} < 0$, and softening ($T > 1$) moves it toward the threshold. Softening thus trades false alarms for recovered misses — the opposite of the sharpening in-domain calibration (Guo et al., 2017) prescribes — and is worth doing precisely when top-1 accuracy is low, which is the out-of-domain regime. Finally, $\mathbf{b}$ is the only term that can change which class wins, hence the only one that can help accuracy or, if fit on a mismatched distribution, actively harm it. Our final system sets $\mathbf{b} = \mathbf{0}$.

Configurations. **S1** and **S2** are the individual branches, ECAPA-TDNN and the frozen Whisper checkpoint; **S3** is S2 after our augmented adaptation; **S4** fuses S1 and S3 at w=0.80 with calibration fit in domain (T=0.9, $\mathbf{b}$ fitted), as first submitted; **S5**, our final submission, is the same fusion with T=2.0 and $\mathbf{b}$=$\mathbf{0}$. S4 and S5 share identical models and differ only in the decision stage.

4 Experimental Setup

Splits. Both branches train on the full 67,015-utterance training split. The 10,806-utterance validation split serves model selection, fusion-weight selection and calibration fitting; to avoid fitting calibration on the data used to judge it, we partition validation into stratified tune and holdout halves (50/50, seed 42), fit on tune, verify on holdout, then refit on the full split. The 878-utterance blind test is used for inference only, and every test number we report comes from the official scorer.

Preprocessing. Audio is mono 16 kHz. Branch A consumes variable-length waveforms padded within each batch. Branch B pads or trims to 480,000 samples and extracts a 128-bin log-mel spectrogram (400-sample FFT window, 160-sample hop), reusing the mel filterbank shipped inside the checkpoint so our front end matches the one the weights were trained under.

Training and selection. Branch A runs 90,000 steps at batch size 4 with AdamW at 10^{-4}, backbone frozen for 15,000 steps then fine-tuned jointly under a five-phase schedule (Appendix C); 4h47m. For Branch B, Elyadata's recipe of unfreezing the upper 16 encoder layers fits only at batch size 1 on our 8 GB GPU, so we freeze the encoder entirely and train only attention pooling and output layers (26,881 parameters), permitting batch size 8: 8,000 steps, Adam at 10^{-4}, 12h41m. Augmentation combines gain, additive noise, speed perturbation, narrowband filtering and chunk dropout with SpecAugment (Park et al., 2019); validation during training uses clean audio. The fusion weight was chosen by grid scan on validation and confirmed on the holdout half, and temperature and bias by

ID	System	Validation ACC	Validation Cost	Official test ACC	Official test Cost
S1	ECAPA-TDNN	81.09	.103	—	—
S2	Whisper, frozen	93.06	.052	—	—
S3	S2 + augmentation	94.24	.043	—	—
S4	Fusion, in-dom. cal.	**95.58**	**.025**	.2540	.2077
S5	Fusion, T=2, $\mathbf{b}$=$\mathbf{0}$	94.35	.060	**.3918**	**.1469**
Decision stage only, identical models, official scorer:					
	T=1.0, $\mathbf{b}$=$\mathbf{0}$			.3918	.1515
	T=3.0, $\mathbf{b}$=$\mathbf{0}$			.3918	.1518

Table 1: Accuracy (%) on validation and official-scorer results on the blind test. S1–S3 were never submitted individually. S4 is best in domain and worst out of it. FPR/FNR for the four submitted variants: .025/.183 (S4), .033/.119 (T=1), .059/.088 (S5), .077/.075 (T=3).

search on the tune half; the final T=2.0 was instead selected from three official submissions at $T \in \{1, 2, 3\}$, for reasons Section 5 makes clear.

Tools and metrics. PyTorch 2.11.0+cu128, torchaudio 2.11.0, SpeechBrain 1.1.0 (**?**), Transformers 5.14.1, `datasets` 3.5.0, NumPy 2.4.6, scikit-learn 1.9.0, Python 3.11.15, CUDA 12.8, on one NVIDIA RTX 5070 Laptop GPU (8 GB).[2] Accuracy is primary; average cost uses β=1 and threshold LLR ≥ 0, with per-class false-alarm and miss rates averaged over the 20 detectors and summed.

5 Results

Official result. S5 scored **0.3918 accuracy and 0.1469 cost**, 10th of 16 teams on a leaderboard running from 0.29 to 0.56. Every visible team lost most of its in-domain accuracy, consistent with the cross-domain design rather than with one system failing.

The in-domain ranking reverses. Table 1 shows the central problem: S4 is better on validation by both metrics yet worse on the official test by 13.8 accuracy points, and S5 retains only 41.5% of its validation accuracy. Model-side gains behaved similarly — augmentation improved S3 over S2 in domain, 93.06 to 94.24 — with no evidence it helped out of domain.

Fusion weight. The system is insensitive to w near its chosen value. Over $w \in [0.70, 0.90]$ validation accuracy varies by 0.59 points and cost by 0.006, with the accuracy optimum at w=0.85 (94.48%) and the cost optimum at w=0.70 (0.0591); our w=0.80 is within 0.13 points and 0.0008 cost of those. Either branch alone is worse than any fusion in this range (Appendix D).

Bias, not temperature, moved accuracy. The three $\mathbf{b}$=$\mathbf{0}$ rows have accuracy identical to ten decimal places (0.3917995444), confirming Proposition 1(iii) empirically: the entire accuracy gain came from discarding the in-domain bias and none from temperature. Temperature moves only cost, tracing a shallow valley minimised at T=2, with miss rate falling and false-alarm rate rising monotonically across the three, as (ii) predicts.

Errors (validation). S5 reaches macro F1 0.935 against weighted F1 0.943, so the equally-weighted cost metric is penalised more than accuracy suggests. MSA is the outlier: recall 0.671 at precision 0.990. The system almost never labels a dialect as MSA but misses a third of true MSA, scattering it to LIB, BAH and ALG at roughly 7% each. A second cluster is geographic: QAT 0.888 (leaking to EGY, UAE, SUD), KUW 0.909, OMA 0.913; the largest single pair is TUN→LEB, 48 utterances. Crucially, most errors are near misses: in a representative MSA→LIB case the true class ranks second (LIB -1.76, MSA -3.59), and likewise in TUN→LEB (LEB -0.84, TUN -1.83), whereas a typical QAT→EGY case puts the true class at rank 6. Appendix E gives per-class results and the confusion matrix.

Errors (blind test). Without labels we can only inspect the predicted distribution, which is severely concentrated: the five most-predicted classes take 57.1% of the 878 utterances against 25% under a uniform split, with EGY at 144 and MAU at 106, while ALG, KUW, MOR, SUD and YEM receive fewer than 15 each. **ALG is never predicted**, though its validation recall is 0.958. The model is not blind to it: ALG ranks in the top three for 72 utterances and in the top four for 104. On those 104 the winning label is MAU 70 times, TUN 10 and MOR 6 — a Maghreb cluster collapsing onto Mauritanian rather than a random failure — and the median score gap to the winner is 2.09, far too wide for any rank-preserving rescaling to close. This is a far miss in the sense of Section 6, which is why neither calibration nor a post-submission class-weighted fine-tune targeting MSA, ALG and

[2] https://huggingface.co/speechbrain/lang-id-voxlingua107-ecapa, https://huggingface.co/Elyadata/ADI-whisper-ADI20, https://huggingface.co/datasets/UBC-NLP/NADI_2026_ADI20_micro.

MOR (built but never submitted, and a wash on validation at 94.46 vs 94.35) moved ALG beyond a single utterance.

6 Discussion

Two error types dominate, and they respond differently to the decision stage. Near misses — the true dialect ranked second or third, as in MSA→LIB or TUN→LEB — are recoverable by softening, which is why $T{=}2$ cut our miss rate from 0.119 to 0.088. Far misses, where the true class falls to rank 6 as in the Gulf cluster, are beyond any rank-preserving transform and need a better model. The single largest weakness, MSA recall of 0.671 at precision 0.990, is of the first kind and also the most interpretable: formal or newsreading register makes speakers of any dialect sound MSA-like, so MSA loses mass to whichever dialect the speaker actually has.

Why does an in-domain bias transfer negatively? The bias encodes per-class threshold corrections tied to the in-domain class balance and confusion structure. Under domain shift those corrections no longer describe the data, but because $\mathbf{b}$ is the one term that moves the $\arg\max$, a stale correction does not simply fail — it relabels utterances that the uncalibrated scores had right. Temperature, being rank-preserving, cannot do this damage, which makes it the safer instrument under uncertainty about the target domain. The rule we would follow next time is to fit calibration on data that is deliberately *not* in-domain: our augmentation pipeline already simulates channel shift, and holding out an augmented calibration set approximates the test condition cheaply. In-domain validation was otherwise an unreliable guide throughout — it ranked S4 above S5 — so every decision after the first submission was driven by official scorer feedback rather than local numbers.

7 Conclusion

We described a two-branch ensemble for NADI 2026 Subtask 2 that scored 0.3918 accuracy and 0.1469 cost on the official blind test, 10th of 16 teams. Our main result is negative and, we think, transferable: calibration parameters fit on in-domain data damaged out-of-domain accuracy, and removing them recovered 13.8 points without touching either model. Knowing which score transformations the evaluation metric is invariant to told us in advance which experiments were worth a submission. For a task defined by domain shift, the decision layer deserves as much attention as the encoder.

Limitations

Scope of the calibration result. Removing the in-domain bias recovers 13.8 of the roughly 69 accuracy points separating our in-domain validation score from our first official submission. Some 55 points remain, so calibration accounts for about a fifth of the domain-shift gap and the majority of it is not explained by anything we identify here. Our claim is that the decision stage is a large and easily overlooked term, not that it is the dominant one.

Test-informed model selection. The final temperature $T{=}2.0$ was chosen by comparing three official submissions rather than by any held-out procedure of our own, and our development after the first submission was guided by official scorer feedback. The blind test therefore participated indirectly in selecting this parameter, and our reported cost should be read accordingly. Accuracy is unaffected by this concern, since Proposition 1(iii) makes it invariant to T.

Unverified out-of-domain gains. S2 and S3 were never submitted individually, so while augmentation improves the Whisper branch in domain, we have no blind-test evidence that it contributed to cross-domain robustness. The same limitation applies to the fusion itself.

Experimental scale. Results come from a single run of each configuration on one 8 GB laptop GPU, with no seed variance reported. The VRAM ceiling prevented fine-tuning the Whisper encoder itself, so Branch B is adapted only at its pooling and output layers.

Unlabelled test data. The blind test has no public labels, so our out-of-domain error analysis rests on the predicted distribution alone and cannot establish why ALG is never predicted: a genuine absence of Algerian speech and a systematic failure of the system look identical from where we stand. Finally, the `ADI-whisper-ADI20` checkpoint carried no stated license at the time of use; we report this rather than assume permissive terms.

Ethics Statement

The corpora used here consist of publicly broadcast speech collected from YouTube by the ADI-17 and

ADI-20 authors, and we introduce no new recordings and no speaker-identifying annotation. Dialect identification nonetheless infers a speaker's likely region of origin from their voice, which carries a risk of profiling if deployed outside research contexts, and errors are not distributed evenly: our system's weakest class, MSA, is the register speakers adopt in formal settings, and its Maghreb confusions are systematic rather than random. We therefore regard the system as suitable for research on dialectal speech technology and for routing within speech pipelines, not for any decision about an individual speaker.

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A Inference Pseudocode

Algorithm 1 Inference for one utterance

Require: waveform x at 16 kHz; weight w; temperature T
$x_B \leftarrow \textsc{PadOrTrim}(x, 30\text{s})$
$\mathbf{m} \leftarrow \textsc{LogMel}(x_B)$
$\mathbf{h} \leftarrow \textsc{WhisperEnc}(\mathbf{m})$
$\mathbf{z}^B \leftarrow \textsc{Head}(\textsc{AttnPool}(\mathbf{h}))$
$\mathbf{z}^B \leftarrow \textsc{Permute}(\mathbf{z}^B)$ ▷ to task label order
$\mathbf{z}^A \leftarrow \textsc{EcapaClassifier}(x)$
$\mathbf{s} \leftarrow w \log \sigma(\mathbf{z}^B) + (1-w) \log \sigma(\mathbf{z}^A)$
$\tilde{\mathbf{s}} \leftarrow \mathbf{s}/T$ ▷ no per-class bias
return $\arg\max_i \tilde{s}_i$ and $\tilde{\mathbf{s}}$

B Proofs

Throughout, $\mathbf{s} \in \mathbb{R}^C$ with $C = 20$ and LLR_i is as in Eq. 1.

(i) Shift invariance. Let $\mathbf{s}' = \mathbf{s} + c\mathbf{1}$. For any $i \neq j$, $s'_j - s'_i = (s_j + c) - (s_i + c) = s_j - s_i$, so every term of the sum is unchanged and so is LLR_i for every i. Since the detection rule depends on $\mathbf{s}$ only through LLR, and $\arg\max_j s'_j = \arg\max_j s_j$, accuracy, the per-class false alarm and miss rates, and the average cost are all identical. □

We verified this empirically before spending a submission on it, sweeping a uniform shift over $[0, 4.0]$: accuracy, false alarm rate, miss rate and cost were bit-identical at every value.

(ii) The top class is always detected. Let $i^\star = \arg\max_j s_j$ and $T > 0$. For every $j \neq i^\star$, $s_j - s_{i^\star} \leq 0$, hence $(s_j - s_{i^\star})/T \leq 0$ and $\exp\big((s_j - s_{i^\star})/T\big) \leq 1$. Averaging $C - 1$ terms each at most 1 gives

$$\frac{1}{C-1} \sum_{j \neq i^\star} \exp\Big(\frac{s_j - s_{i^\star}}{T}\Big) \leq 1,$$

and since $-\log u \geq 0$ for $u \in (0, 1]$, $\mathrm{LLR}_{i^\star}(\mathbf{s}/T) \geq 0$ at every positive temperature. □

(iii) Temperature preserves ranking. For $T > 0$ the map $u \mapsto u/T$ is strictly increasing, so it preserves the order of the entries of $\mathbf{s}$ and in particular the $\arg\max$; accuracy is invariant to T. For the limit, $(s_j - s_i)/T \to 0$ as $T \to \infty$, so each $\exp\big((s_j - s_i)/T\big) \to 1$, the average of the $C - 1$ terms tends to 1, and $\mathrm{LLR}_i(\mathbf{s}/T) \to -\log 1 = 0$. □

Remark. By (ii), every miss originates from an utterance whose true dialect is not ranked first, where the true class may have $\mathrm{LLR} < 0$. By (iii), increasing T drives every ratio toward zero, so classes near the threshold can cross it. We stress that (ii) and (iii) do not establish that raising T can only *add* detections. What they do give is the limiting behaviour: writing $d_j = s_j - s_i$ and expanding for large T,

$$\mathrm{LLR}_i(\mathbf{s}/T) = -\frac{1}{T} \cdot \frac{1}{C-1} \sum_{j \neq i} d_j + O(T^{-2}),$$

so as T grows class i is detected precisely when its mean gap $\frac{1}{C-1}\sum_{j \neq i} d_j$ is negative. We make no monotonicity claim for intermediate T, and the direction of the trade is therefore an empirical matter. On the official test it went one way at every point measured: as T went $1.0 \to 2.0 \to 3.0$, the miss rate fell ($0.119 \to 0.088 \to 0.075$) and the false alarm rate rose ($0.033 \to 0.059 \to 0.077$), while accuracy was bit-identical. The trade is worth making when top-1 accuracy is low.

C Full Hyperparameters

Branch A (ECAPA-TDNN). 90,000 steps, batch size 4, AdamW, learning rate 10^{-4}, cross-entropy

loss, shuffle buffer 1000 with seed 42. The backbone is frozen for the first 15,000 steps while only the new 20-way head trains, then the whole model is unfrozen. The learning-rate schedule chains five phases: a linear ramp from $\frac{1}{3}$ to 1 over the first 9,000 steps; a constant phase to step 15,000; a second linear ramp from $\frac{1}{3}$ to 1 over 9,000 steps beginning at the unfreeze point; a further constant phase to step 30,000; and exponential decay with $\gamma = 0.9999$ from step 60,000 onward.

Branch B (attention pooling + head). All 32 Whisper encoder layers frozen; 26,881 trainable parameters; 8,000 steps, batch size 8, Adam, learning rate 10^{-4}. Evaluation every 500 steps on clean (unaugmented) validation audio.

Augmentation. Applied to training audio only, each component sampled independently per utterance:

- random gain, $\mathcal{U}(-6, 6)$ dB, $p = 0.5$;
- additive white Gaussian noise at SNR $\mathcal{U}(0, 20)$ dB, $p = 0.6$;
- speed perturbation by resampling, factor drawn from $\{0.9, 0.95, 1.05, 1.1\}$, $p = 0.4$;
- narrowband (telephone-channel) simulation via a bandpass biquad with centre frequency 1700 Hz and $Q = 0.35$, $p = 0.35$;
- chunk dropout, zeroing one contiguous span covering 2–10% of the clip, $p = 0.3$;
- SpecAugment (Park et al., 2019) on the log-mel spectrogram: two frequency masks of up to 27 bins and two time masks of up to 100 frames.

Calibration search. Temperature searched over $[0.5, 3.0]$ in steps of 0.1; per-class bias fit by three rounds of coordinate descent, each sweeping one class at a time over $[-2.0, 2.0]$ in steps of 0.2 while holding the others fixed. Both minimise average cost on the tune half of validation.

D Fusion-Weight Sensitivity

Table 2 sweeps the fusion weight w of Section 3 on the full validation split, with the final decision stage applied (T=2.0, $\mathbf{b}$=$\mathbf{0}$). All numbers are recomputed from saved per-branch logits, so no model is re-run. The endpoints w=0 and w=1 are the single branches.

w	ACC (%)	Cost	FPR	FNR
0.00	81.09	.0942	.0507	.0435
0.50	91.09	.0632	.0510	.0122
0.60	92.84	.0612	.0507	.0105
0.65	93.46	.0597	.0507	.0090
0.70	93.89	**.0591**	.0506	.0085
0.75	94.14	.0592	.0508	.0084
0.80	94.35	.0599	.0517	.0082
0.85	**94.48**	.0616	.0527	.0089
0.90	**94.48**	.0651	.0543	.0108
0.95	94.40	.0683	.0566	.0117
1.00	94.24	.0736	.0609	.0127

Table 2: Fusion-weight sweep on validation at T=2.0, $\mathbf{b}$=$\mathbf{0}$. The submitted system uses w=0.80. Accuracy peaks at w=0.85 and cost at w=0.70, but the surface is flat between them.

E Detailed Error Analysis

Figure 1 shows the row-normalised confusion matrix for the final system S5 on the validation split, and Table 3 the corresponding per-class precision, recall, F1 and support.

Dialect	**P**	**R**	**F1**	**Sup.**
MSA	0.990	0.671	0.800	283
BAH	0.919	0.997	0.956	317
TUN	0.992	0.935	0.962	1649
ALG	0.878	0.958	0.916	240
EGY	0.916	0.960	0.938	650
IRA	0.966	0.987	0.976	628
JOR	0.968	0.921	0.944	367
KSA	0.929	0.965	0.947	368
KUW	0.985	0.909	0.946	430
LEB	0.812	0.979	0.888	383
LIB	0.951	0.928	0.939	627
MAU	0.934	0.995	0.964	213
MOR	0.949	0.974	0.961	380
OMA	0.925	0.913	0.919	632
PAL	0.936	0.993	0.964	430
QAT	0.979	0.888	0.932	914
SUD	0.837	0.990	0.907	197
SYR	0.897	0.953	0.924	446
UAE	0.955	0.981	0.968	1130
YEM	0.946	0.971	0.958	522
Macro F1 0.935		Weighted F1 0.943		10806

Table 3: Per-class results for S5 on the validation split.

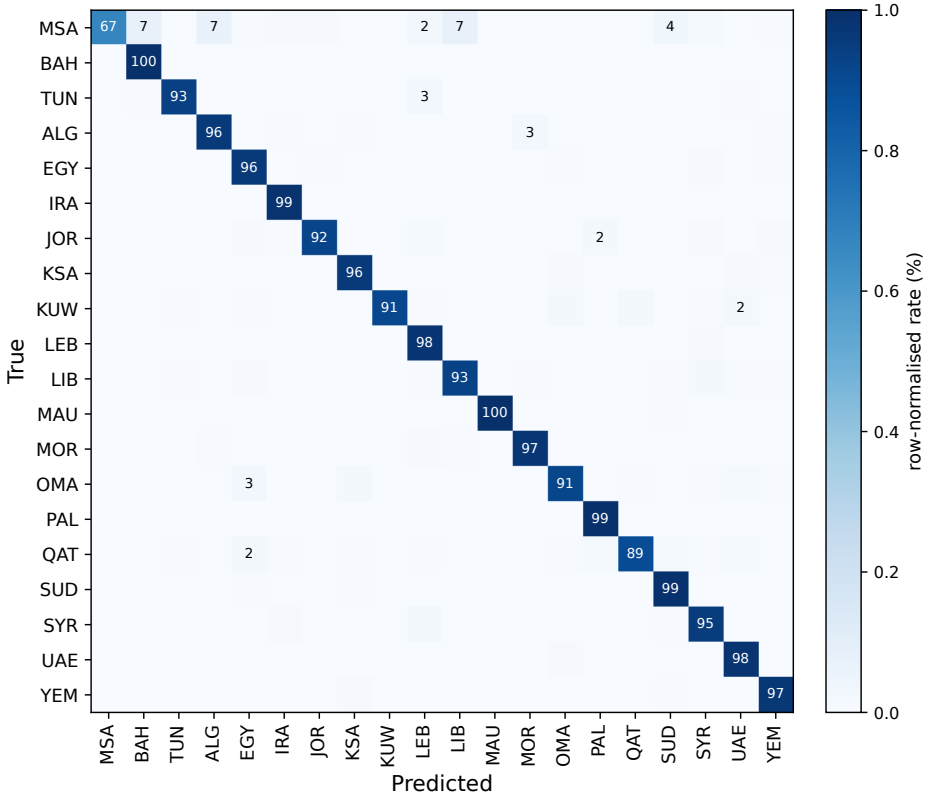


Figure 1: Row-normalised confusion matrix for S5 on the validation split. Cells below 2% are left unlabelled. The MSA row is the visibly weakest, dispersing across LIB, BAH and ALG.